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Effects of biochar combined with different nitrogen fertilization rates on turf quality and carbon footprint in a turf system

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ABSTRACT

Biochar has been widely recognized for its potential to mitigate greenhouse gas emissions and enhance soil fertility through increased soil organic carbon (SOC) storage; however, systematic evaluations of carbon balance and ecological benefits in turf systems remain scarce. This study included eight treatments combining four N (nitrogen) rates (N_0: 0, N_1: 150, N_2: 225, N_3: 300) kg N ha^{-1} with two biochar practices (SB_0: 0, SB:40) t ha^{-1}. The results showed that biochar application treatments (N_0SB, N_1SB, N_2SB, and N_3SB) reduced cumulative fluxes of soil N_2O and CH_4 (global warming potential, GWP) by 17.37 %, 26.62 %, 32.73 %, and 30.97 %, net global warming potential (NGWP) 43.71 %, 369.64 %, 1046.17 %, and 545.83 %, and CF by 32.01 %, 192.81 %, 356.66 %, and 253.34 %, while increasing SOC sequestration by 129.20 %, 138.61 %, 133.89 %, and 133.61 %, total biomass by 3.99 %, 14.22 %, 15.51 %, and 14.78 %, and turf normalized difference vegetation index (NDVI) by −0.18 %, 2.81 %, 2.81 %, and 1.89 % compared to non-biochar treatments. Although biochar slightly increased ecosystem respiration, the effect was not significant ($p > 0.05$). Notably, compared to N_3SB, N_2SB significantly reduced N_2O by 12.85 %, CH_4 by 11.47 %, CF (carbon footprint) by 16.01 %, and NGWP by 9.82 %, while ecosystem respiration, SOC storage, turfgrass total biomass, and NDVI exhibited no significant differences between the two treatments. Overall, the co-application of biochar with 225 kg N ha^{-1} is a win−win strategy for reducing CF, enhancing carbon sequestration, and maintaining turf quality.

1. Introduction

Climate change, driven by anthropogenic increases in greenhouse gas (GHG) concentrations, is widely recognized as a significant threat to global ecosystems. The rise in GHG emissions is primarily due to human activities, particularly those linked to agricultural production and urban landscaping (Lu et al., 2015; Dutt and Tanwar, 2020; Gills and Morgan, 2020). Substantial research has been conducted to improve soil carbon (C) sinks and reduce GHG emissions in agricultural and natural ecosystems (Adewale et al., 2019; Bai et al., 2023). Studies identified higher nitrogen (N) fertilizer application rates as key factors contributing to increased soil GHG emissions in agricultural ecosystems (Yao et al., 2020; Yu et al., 2023). However, relevant research has rarely been conducted in urban landscaping systems, particularly in urban turf systems. Similar to agricultural ecosystems, turf systems are often subject to extensive fertilization and intensive management, resulting in substantial GHG emissions (Horgan et al., 2002; Townsend-Small and Czimczik,

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2010). This gap creates uncertainty regarding efforts to mitigate global warming and environmental degradation. Therefore, effective management practices for urban turf systems to enhance C sequestration and reduce GHG emissions are essential for climate change mitigation.

Turf systems are essential components of urban landscapes, commonly found in sports fields, parks, golf courses, and residential yards (Yang et al., 2019). Turf systems provide various environmental and human health benefits, such as green recreational spaces, increased carbon sequestration, and minimized nutrient loss from soils (Golubiewski, 2006; Monteiro, 2017). Although these environmental processes likely influence the biogeochemical cycles of C and N, urban turf systems are frequently omitted from regional and global GHG inventories due to the assumption that urban land areas are too small to contribute meaningfully to biogeochemical GHG fluxes. Moreover, urban ecosystems are largely excluded from IPCC climate scenarios due to limited biogeochemical data (IPCC, 2006, 2013). However, turf covers more land area in the U.S. than any irrigated crop (Milesi et al., 2005), and its expansion is projected to continue with urbanization in China (Zhan et al., 2023). Fertilization, critical for maintaining turf aesthetics, also promotes optimal conditions for enhanced soil N_2O emissions (Gu et al., 2015). Studies have shown that N_2O emissions from fertilized urban turf soils can equal or exceed those from agricultural soils (Hall et al., 2008; Townsend-Small et al., 2011; Braun and Bremer, 2018; Chen et al., 2018). For example, Kaye et al. (2004) reported that turf, occupying only 6.4 % of total land area, contribute 30 % of the total soil N_2O emissions in certain regions. A meta-analysis further confirmed that fertilization increases soil N_2O emissions by 41 % in turf systems compared to unfertilized plots (Dutt and Tanwar, 2020). These findings align with agricultural soil studies (Linquist et al., 2012), indicating that urban turf systems under high fertilizer inputs are significant sources of atmospheric N_2O (Bremer, 2006; Li et al., 2013), with potential global climate implications.

The application of biochar in agricultural soils is recognized as a win-win strategy for climate change mitigation and crop productivity enhancement (Zhang et al., 2024b). Biochar, a stable carbon-rich material produced via biomass pyrolysis, improves soil properties through its high porosity, adsorption capacity, and functional groups resistant to microbial degradation (Roberts et al., 2010). Studies have shown that the complex aromatic structure and inherent chemical inertness of biochar provide high chemical and microbial stability in soils, effectively reducing soil organic carbon (SOC) decomposition (Aliyu et al., 2019; Hu et al., 2024). Furthermore, its high adsorption capacity for inorganic nitrogen reduces substrate availability for GHG emissions, thereby decreasing emissions (Shen et al., 2014; Lehmann et al., 2021). Biochar also enhances nutrient availability, promotes crop growth, increases SOC input (Major et al., 2010; Woolf et al., 2010; Zheng et al., 2017; Zhang et al., 2024b), improves soil structure (Yanai et al., 2007), and stimulates microbial activity (Steinbeiss et al., 2009). Moreover, bio-oil and biogas produced during biochar production can be harnessed for electricity generation, offsetting some GHG and reactive nitrogen emissions while reducing environmental costs and enhancing economic benefits (Ji et al., 2018; Xia et al., 2023). Despite these benefits, research on biochar in turf ecosystems remains scarce, with limited understanding of its effects on carbon balance, environmental efficiency, and turf growth (Khan and Shea, 2013; Tian et al., 2019). Further investigations are needed to evaluate biochar's role in carbon cycling, SOC sequestration, and GHG mitigation within urban turf systems.

The carbon footprint (CF) quantifies the total direct or indirect CO_2 equivalent emissions from all activities within a production system (Adewale et al., 2019). It is widely used to assess the GHG emissions resulting from production activities throughout their lifecycle (Huang et al., 2019; Wang et al., 2019c). In agricultural systems, CF analysis has been applied to evaluate environmental impacts, promote low-carbon practices (Yadav et al., 2018; Wang et al., 2019a), and quantify carbon sequestration (Braun and Bremer, 2019; Sapkota et al., 2020; Wang et al., 2022b), and direct GHG emissions (Brandani et al., 2021; Law et al., 2021). For turf ecosystems, the GHG budget includes direct emissions (e.g., N_2O from soils) and indirect emissions associated with management activities, such as fuel consumption by machinery, energy use for irrigation, and the production and application of fertilizers and pesticides (Bekken and Soldat, 2021). Therefore, quantifying CF in turf systems is essential to determine whether they function as net GHG sinks or sources.

Given the above, we hypothesized that biochar application would enhance turf growth and SOC sequestration while reducing GHG emissions, with effects varying across nitrogen input levels. The objectives of this study were to evaluate the effects of biochar application by: 1) determining the impact of biochar on carbon emissions, turfgrass growth, and SOC storage; 2) analyzing carbon balance and CF; and 3) providing recommendations for sustainable turf ecosystem management.

2. Materials and methods

2.1. Experimental site

The study was conducted over a two-year period in a hybrid bermudagrass (*Cynodon transvaalensis* × *Cynodon dactylon*,'Tifdwarf') turf system at the Jurong Turfgrass Science Experimental Station of Nanjing Agricultural University (31°49′N, 119°14′E). The hybrid bermudagrass turf had been established for four years before the trial. The experimental site has a subtropical monsoon climate, with an average annual temperature of 16.6°C and annual precipitation of 1192 mm. The soil type is classified as sandy clay loam. The topsoil (0–10 cm) had the following properties: pH 7.71, bulk density (BD) 1.43 g cm^{-3}, total organic carbon 3.85 g kg^{-1}, total nitrogen 0.95 g kg^{-1}, and total phosphorus 0.69 g kg^{-1}.

2.2. Experimental design and turf management

The experiment was conducted using a randomized block design from April 2022 to April 2024, and comprised eight treatments, each with three replicates (3 m × 3 m). The treatments included two biochar application rates (SB_0: no biochar, SB: 40 t ha^{-1}) and four N levels (N_0: no N fertilizer, N_1: 150 kg N ha^{-1}, N_2: 225 kg N ha^{-1}, N_3: 300 kg N ha^{-1}). Hybrid bermudagrass stems were mechanically

cut with a rotary blade, loaded into a seed drill and evenly spread across the experimental plots. Biochar was thoroughly incorporated into the top 0–15 cm of soil only at the start of the experiment and not reapplied thereafter. The biochar used in this study, purchased from Zhenjiang Zedi Agricultural Biotechnology Co. Ltd (Zhenjiang, China), was produced from rice straw through pyrolysis at 500–600°C. Its physicochemical properties were: pH 8.95, total organic carbon 650.16 g kg^{-1}, total N 9.25 g kg^{-1}. N fertilizer (urea, N = 46.0 %) was applied in three equal splits in April, July, and September of each year. All plots only received uniform phosphorus (100 kg P_2O_5 ha^{-1} as superphosphate, P_2O_5 = 12.0 %) and potassium (50 kg K_2O ha^{-1} as potassium chloride, K_2O = 60.0 %) annually in April. After turf establishment, plots were mowed twice monthly to maintain a height of 4 cm. Clippings and dormant aboveground biomass were removed from the plots to simulate typical turf management practices. Detailed management practices and input parameters are provided in Tables S1 and S2, and irrigation schedules are described in Table S3.

2.3. GHG sampling and measurements

The static closed-chamber method was employed to measure soil N_2O and CH_4 fluxes, and ecosystem respiration (Re) fluxes throughout the study. Prior to the experiment, a PVC base (10 cm× 50 cm× 50 cm) was buried 10 cm into the soil and remained in place for the entire study. Each base had a groove along its upper edge to allow for water filling, which sealed the sampling chamber during gas collection. The sampling chamber was equipped with a circulating fan to ensure thorough gas mixing and an electronic to monitor the internal temperature. Additionally, the chamber exterior was wrapped with sponge and aluminum foil layers to minimize temperature fluctuations during sampling. After sealing the chamber, gas samples were collected with a 50 mL syringe at 0, 10, 20, and 30 min after sealing the chamber. Sampling frequency was intensified to three times per week following fertilization, while monthly measurements were conducted for soil N_2O, CH_4, and Re fluxes during the non-fertilization period. All gas sampling collection was conducted between 09:00 and 11:00 local time. Gas concentrations of N_2O, CH_4, and CO_2 were analyzed within 48 h using a gas chromatograph (Shimadzu GC-2010 Plus, Kyoto, Japan). The specific formulas for greenhouse gases were calculated as follows (Qian et al., 2022):

$$F = \rho \times \frac{V}{A} \times \frac{\Delta c}{\Delta t} \times \frac{273}{273 + T}$$

Where F represents the fluxes of N_2O (mg N_2O m^{-2} h^{-1}), CH_4 (mg CH_4 m^{-2} h^{-1}), or Re (mg CO_2 m^{-2} h^{-1}), ρ denotes the standard gas density (mg cm^{-3}), V and A are the volume (m^3) and bottom area (m^2) of the chamber, respectively. Δc/Δt is the change in gas concentration in the chamber, and T is the temperature inside the chamber during the measurement (℃).

The annual cumulative GHG fluxes were calculated using the following formula (Qian et al., 2022):

$$E = \sum_{i}^{n} \frac{F_i + F_{i+1}}{2} \times (t_{i+1} - t_i) \times 24$$

Where E is the annual cumulative soil N_2O and CH_4, and cumulative Re fluxes (kg N_2O ha^{-1}, kg CH_4 ha^{-1}, and kg CO_2 ha^{-1}), t_i and t_{i+1} represent two consecutive sample collections, and t represents the sample collection days.

Global warming potential (GWP) (kg CO_2-eq ha^{-1}) quantifies the exchange of N_2O and CH_4 expressed as CO_2 equivalents, between the soil and atmosphere. Net global warming potential (NGWP) was calculated based on observation of SOC sequestration rate, and the budgets of N_2O and CH_4 were converted to CO_2 equivalents based on 100-year radiation intensity. GWP and NGWP were calculated as (Sun et al., 2022; Chen et al., 2023):

$$GWP = 298 \times E_{N_2O} + 34 \times E_{CH_4}$$

$$NGWP = 298 \times E_{N_2O} + 34 \times E_{CH_4} - \Delta SOC \times \frac{44}{12}$$

Where E_{N_2O} and E_{CH_4} are the cumulative emissions of turf GHG (kg ha^{-1}), 298 and 34 denote the GHG conversion equivalents, ΔSOC is the annual SOC variation in 0–15 cm soil depth (Mg C ha^{-1} yr^{-1}), and 44/12 is the factor for C conversion into CO_2.

2.4. Soil sampling and measurements

Soil water content (SWC) and soil temperature were measured concurrently with gas sampling at a 7.6 cm depth using a Fieldscout TDR-350 (Spectrum Technologies Inc., Aurora, IL, USA). Soil water-filled pore space (WFPS) was then calculated by dividing the product of SWC and soil bulk density (BD) by the total soil porosity (Bai et al., 2021):

$$\mathrm{WFPS}(\%) = \frac{\mathrm{SWC} \times \mathrm{BD}}{1 - \frac{\mathrm{BD}}{2.65}} \times 100$$

Soil samples (0–15 cm) were collected annually in April from 2022 to 2024. BD was determined via the ring knife method. Soil pH (soil:water 1:5, w/v) was determined with a portable pH meter (FieldScout pH400, Spectrum Technologies Inc., Aurora, IL, USA). Soil from four random points per plot was homogenized, air-dried, sieved through a 0.25 mm mesh, and subjected to SOC analysis. To remove inorganic carbon, soil samples were moistened with ultrapure water, fumigated with HCl under vacuum for 48 h, and analyzed

with a Multi N/C 2100 (Analytik Jena, Germany) (Wang et al., 2023). SOC storage was calculated based on the relationship between soil bulk density, soil thickness and soil organic carbon content (Zhang et al., 2021). The SOC storage was calculated using the following equation(Xu et al., 2019):

$$SOC_{storage} = SOC \times BD \times H$$

Where $SOC_{storage}$ is the amount of SOC sequestered in the soil (Mg C ha^{-1}); BD represents the bulk density of the soil (g cm^{-3}) and H represents the thickness of the soil (0–15 cm).

The SOC storage plays a significant role in determining the CF. When ΔSOC is positive, it indicates SOC sequestration; when ΔSOC is negative, it signifies a loss from the soil carbon pool. The annual SOC storage change was calculated as follows (Li et al., 2021):

$$\Delta SOC = \frac{S_{2022} - S_{2024}}{2}$$

Where ΔSOC represents the annual change in SOC storage (Mg C ha^{-1} yr^{-1}), S_{2022} and S_{2024} denote the organic carbon storage at the start (2022) and end (2024) of the experiment, respectively, and 2 is the number of years the experiment lasted.

2.5. Measurements of biomass and NDVI

The normalized difference vegetation index (NDVI) reflects the relationship between chlorophyll absorption in the visible spectrum and reflection in the near-infrared spectrum. Higher values indicate greater chlorophyll content, denser canopies, and a greener appearance, making it a valuable tool for assessing turfgrass health and quality (Trenholm et al., 1999; Bell et al., 2002). From April to November each year during the experiment, NDVI was measured monthly before mowing using a FieldScout CM1000 NDVI meter (Spectrum Technologies Inc., Aurora, IL, USA). Six random points per plot were recorded. The annual mean NDVI of turfgrass was used for subsequent analysis.The sum of clippings and dormant aboveground biomass, dried at 65℃ to a constant weight, was used as aboveground biomass. Root biomass was determined by collecting three soil cores (30 cm depth × 10 cm diameter) per plot in November each year, followed by cleaning, drying, and weighing.

2.6. Carbon footprint

CF, expressed as CO_2 equivalents (CO_2-eq), quantifies total greenhouse gas emissions from a production system (Xue et al., 2016). In this study, the system boundary encompassed all annual turf maintenance activities. Following IPCC (2006) guidelines, CO_2 emissions were excluded due to dynamic equilibrium between photosynthesis and respiration. Emission factors for inputs (Table 1) were sourced from published literature. The estimation formula for CF during the turf maintenance process was as follows (Xiao et al., 2024):

$$CF = \sum_i^n (AI_i \times EF_i) + E_{N_2O} + E_{CH_4} - \Delta SOC \times \frac{44}{12}$$

where, AI_i represents the input during the turf maintenance management process (Table S1, Table S2); EF_i represents the emission factor of each agricultural input (Table 1); E_{N_2O} denotes the cumulative soil N_2O emissions (kg CO_2-eq ha^{-1}); E_{CH_4} denotes the cumulative soil CH_4 emissions (kg CO_2-eq ha^{-1}). ΔSOC is the annual SOC change in the 0–15 cm soil layer (Mg C ha^{-1} yr^{-1}), and 44/12 is conversion factor from carbon to CO_2.

2.7. Statistical analysis

All data were processed, analyzed, and plotted using SPSS 18.0 (SPSS Inc., Chicago, IL, USA), and Origin 2022 software (OriginLab Inc., USA). The data was analyzed by one−way variance to determine the significant differences among treatments, followed by Duncan's multiple−range test at $p \leq 0.05$.

Table 1
Emission factors of turf management inputs used for Carbon footprint accounting in the study.

Emission items	Unit	Emission factor kg CO_2-eq $unit^{-1}$	Reference
Urea	kg N	3.44	(Chen et al., 2015)
P_2O_5	kg P_2O_5	0.4	(Chen et al., 2015)
K_2O	kg K_2O	0.3	(Chen et al., 2015)
Insecticide	kg	16.14	(Wang et al., 2016)
Diesel	L	3.75	(Chen et al., 2014)
Electricity	kwh	1.14	(Zhang et al., 2016)
Biochar	t	−0.70	(Liu et al., 2022)

3. Results

3.1. GHG emissions

Soil N_2O fluxes under all treatments exhibited a multi-peak pattern, with primary peaks observed on the fourth day post-fertilization, followed by a gradual decline to lower levels (Fig. 1a). Over the two-year period, both peak and cumulative soil N_2O fluxes showed a clear upward trend with increasing N fertilizer application, as N fertilizer stimulated emissions, whereas biochar

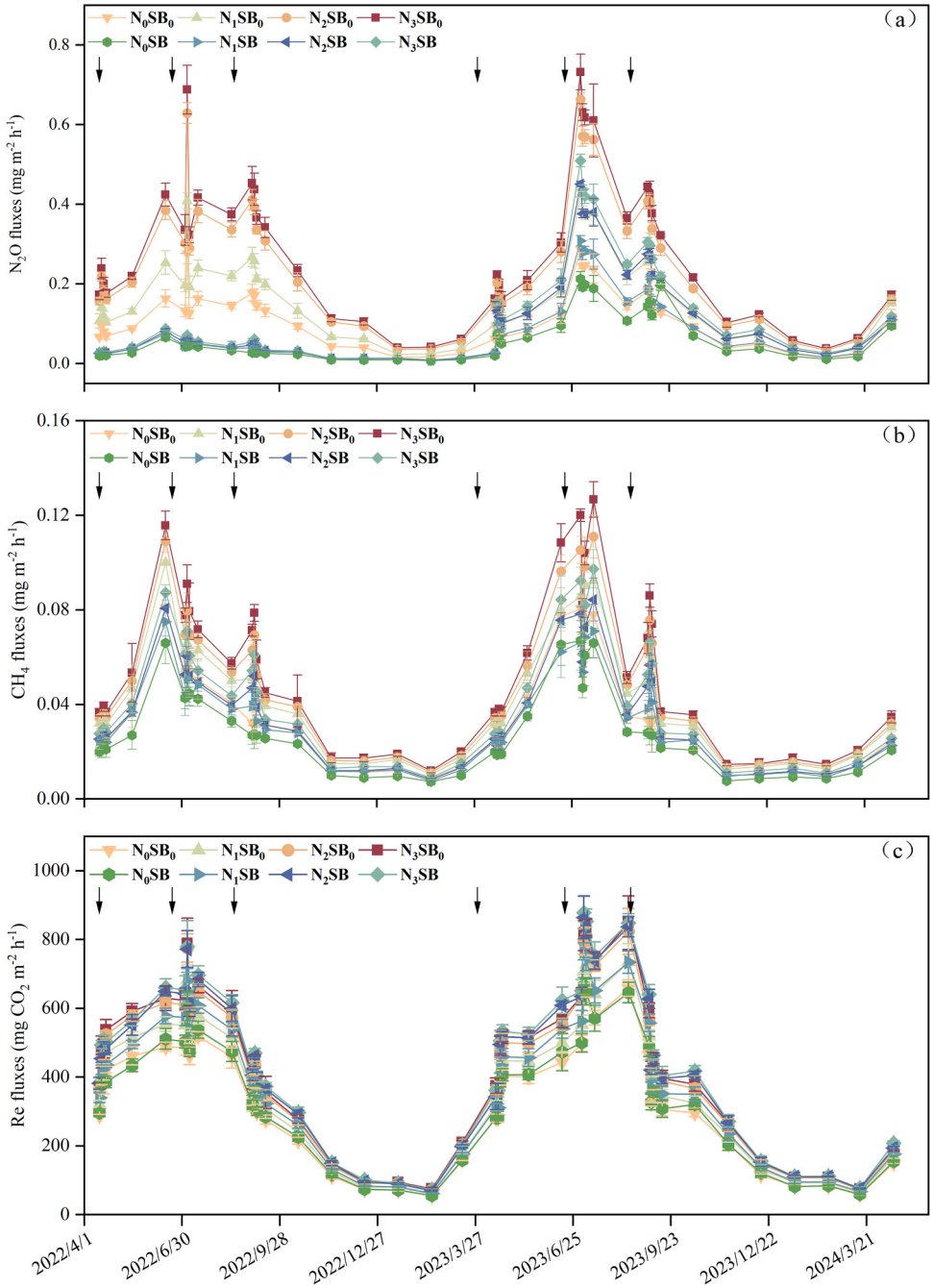


Fig. 1. Soil N_2O and CH_4 fluxes, and ecosystem respiration (Re) fluxes (mg m^{-2} h^{-1}) under different treatments from April 2022 to April 2024. N_0SB_0: no N fertilization and biochar; N_1SB_0: 150 kg N ha^{-1} with no biochar; N_2SB_0: 225 kg N ha^{-1} with no biochar; N_3SB_0: 300 kg N ha^{-1} with no biochar; N_0SB: 40 t ha^{-1} biochar with no N fertilization; N_1SB: 150 kg N ha^{-1} with 40 t ha^{-1} biochar; and N_2SB: 225 kg N ha^{-1} with 40 t ha^{-1} biochar; N_3SB: 300 kg N ha^{-1} with 40 t ha^{-1} biochar. Black arrows indicate the fertilization dates.

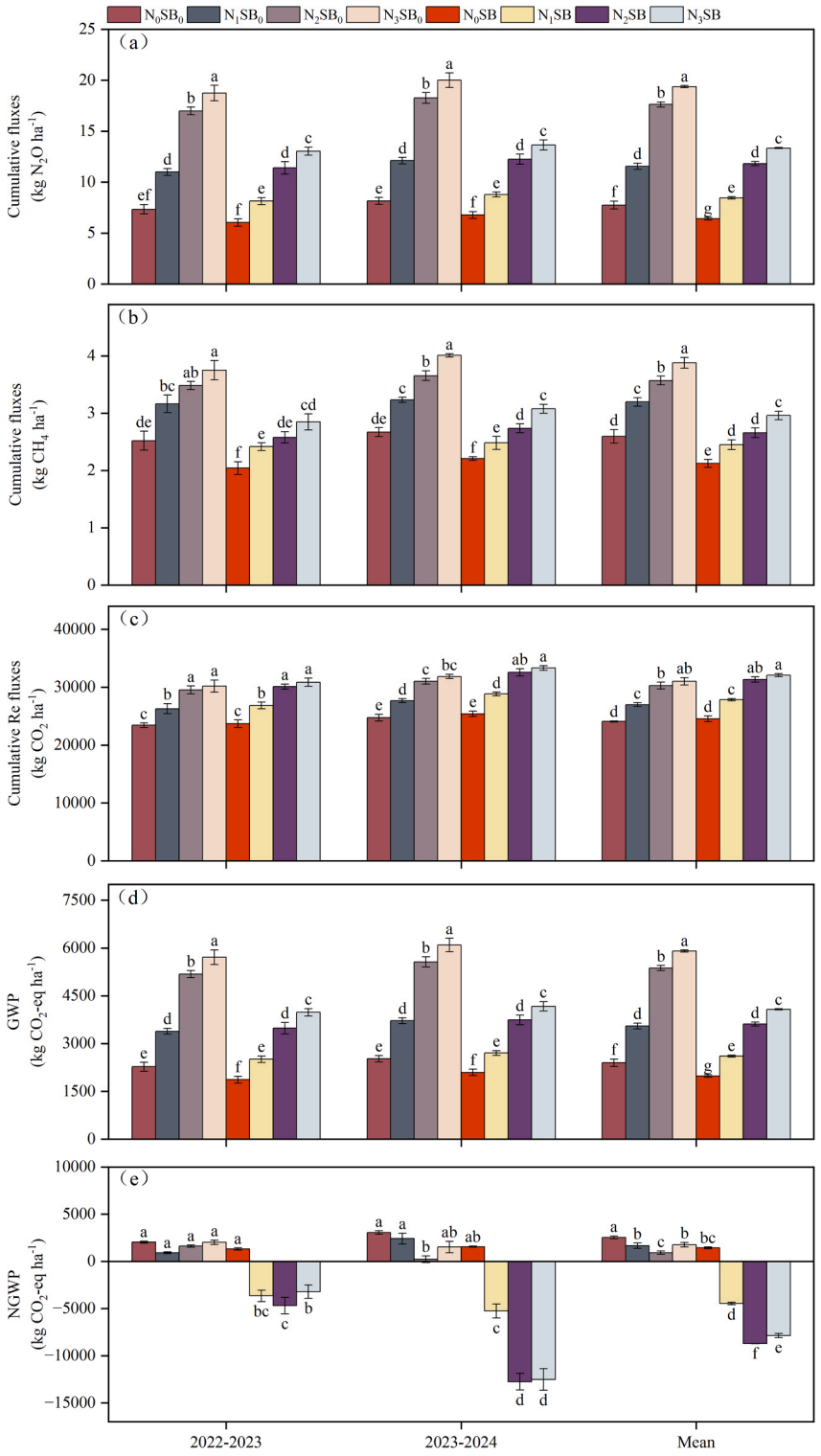


Fig. 2. The cumulative fluxes of soil N_2O (kg N_2O ha^{-1}) (a), soil CH_4 (kg CH_4 ha^{-1}) (b), ecosystem respiration (Re) (kg CO_2 ha^{-1}) (c), global warming potential (GWP) (kg CO_2-eq ha^{-1}) (d), and net global warming potential (NGWP) (kg CO_2-eq ha^{-1}) (e) under different treatments from April 2022 to April 2024. N_0SB_0: no N fertilization and biochar; N_1SB_0: 150 kg N ha^{-1} with no biochar; N_2SB_0: 225 kg N ha^{-1} with no biochar; N_3SB_0: 300 kg N ha^{-1} with no biochar; N_0SB: 40 t ha^{-1} biochar with no N fertilization; N_1SB: 150 kg N ha^{-1} with 40 t ha^{-1} biochar; and N_2SB: 225 kg N ha^{-1} with 40 t ha^{-1} biochar; N_3SB: 300 kg N ha^{-1} with 40 t ha^{-1} biochar. Error bars denote the standard deviation (n = 3). Lowercase letters indicate differences between different treatments ($p < 0.05$).

amendment mitigated this effect (Figs. 1a, 2a). Soil N_2O fluxes varied significantly by season, peaking in summer and reaching their lowest levels in winter (Fig. 1a). The average cumulative soil N_2O fluxes (kg N_2O ha^{-1}) followed this order: N_0SB (6.41) < N_0SB_0 (7.75) < N_1SB (8.47) < N_1SB_0 (11.55) < N_2SB (11.82) < N_3SB (13.34) < N_2SB_0 (17.62) < N_3SB_0 (19.37). Under the same N fertilizer gradient, biochar application consistently mitigated cumulative soil N_2O fluxes, with average mitigation rates of 17.34 % (N_0SB), 26.72 % (N_1SB), 32.90 % (N_2SB), and 31.13 % (N_3SB) (Fig. 2a).

The dynamics of soil CH_4 fluxes from different treatments fluctuated significantly over time, indicating that the turf soil in this trial served as a minor source of CH_4 (Figs. 1b, 2b). Throughout the study, N fertilizer application stimulated soil CH_4 emissions, whereas biochar amendment significantly alleviated this effect. In treatments without biochar, N_1SB_0, N_2SB_0, and N_3SB_0 showed average soil CH_4 fluxes 25.69 %, 43.46 %, and 57.15 % higher than the N_0SB_0, with cumulative fluxes increasing of 23.27 %, 37.60 %, and 49.55 %, respectively. However, under the same N levels, biochar amendment significantly reduced cumulative soil CH_4 fluxes by 18.15 %–25.56 % (Fig. 2b).

Re fluxes exhibited a multi-peak pattern, with peaks occurring primarily after fertilization (Fig. 1c). N fertilizer significantly increased the average fluxes and cumulative fluxes of Re. Compared to N_0SB_0, the average Re fluxes for N_1SB_0, N_2SB_0, and N_3SB_0 treatments increased by 11.93 %, 25.53 %, and 28.76 %, respectively, over the two years, with corresponding increases in cumulative fluxes of 12.01 %, 25.72 %, and 28.82 % (Figs. 1c, 2c). Under the same N fertilizer gradient, biochar amendment stimulated the Re fluxes to varying degrees, with inter-annual differences. In 2022–2023, the average Re fluxes for treatments with biochar ranged from 328.45 to 428.28 mg m^{-2} h^{-1}, compared to 324.26–417.60 mg m^{-2} h^{-1} in non-biochar treatments, representing an average increased of 2.16 %. In 2023–2024, Re fluxes in biochar-amended treatments ranged from 354.06 to 463.72 mg m^{-2} h^{-1}, compared to 347.24–447.62 mg m^{-2} h^{-1} in non-biochar treatments, with an average increased of 3.25 %. Under the same N fertilizer levels, biochar-amended treatments showed average cumulative Re fluxes increased by 1.16 %–2.14 % in 2022–2023 and 2.63 %–5.00 % in 2023–2024, compared to to non-biochar treatments. Although cumulative Re fluxes increased moderately over the two years (1.91 %–3.48 %), no significant differences were observed ($p > 0.05$) (Fig. 2c).

3.2. GWP and NGWP

In all treatments, soil N_2O was identified as the primary contributor to GWP, with N fertilizer serving as the key factor driving soil N_2O fluxes (Fig. 2d). Compared to N_0SB_0, the average GWP for N_1SB0, N_2SB_0, and N_3SB_0 treatments increased by 48.15 %, 124.09 %, and 146.29 %, respectively, during the two-year study. However, biochar amendment led to significant reductions in GWP (Fig. 2d). Compared to non-biochar treatments, biochar application decreased GWP by 17.37 % (N_0SB), 26.62 % (N_1SB), 32.73 % (N_2SB), and 30.97 % (N_3SB) (Fig. 2d). Interestingly, NGWP initially increased and then decreased with increasing N application in all treatments, regardless of biochar addition (Fig. 2e). However, compared to non-biochar treatments, biochar application significantly reduced NGWP, converting it from positive to negative values. This indicates that biochar could transform the NGWP of turf soil from a "source" to a "sink" (except for N_0SB).

3.3. Effects of different treatments on turf growth and quality

The effects of different treatments on turf growth and quality are summarized in Table 2 and Fig. 3. N fertilizer application

Table 2
The impact of different treatments on aboveground, belowground, and total biomass (t ha^{-1}) from 2022 to 2024.

Year	Treatment	Aboveground biomass	Belowground biomass	Total biomass
2022 −2023	N_0SB_0	2.23 ± 0.21d	0.25 ± 0.02c	2.48 ± 0.22d
	N_1SB_0	5.86 ± 0.12c	0.61 ± 0.01b	6.46 ± 0.13c
	N_2SB_0	6.37 ± 0.12b	0.64 ± 0.02b	7.02 ± 0.12b
	N_3SB_0	6.40 ± 0.13b	0.65 ± 0.02b	7.05 ± 0.14b
	N_0SB	2.29 ± 0.07d	0.26 ± 0.02c	2.54 ± 0.08d
	N_1SB	6.60 ± 0.18b	0.64 ± 0.02b	7.24 ± 0.16b
	N_2SB	7.22 ± 0.25a	0.73 ± 0.02a	7.94 ± 0.27a
	N_3SB	7.20 ± 0.21a	0.74 ± 0.02a	7.93 ± 0.22a
2023 −2024	N_0SB_0	2.29 ± 0.22d	0.27 ± 0.02d	2.56 ± 0.23d
	N_1SB_0	6.16 ± 0.47c	0.63 ± 0.03c	6.79 ± 0.49c
	N_2SB_0	7.15 ± 0.28b	0.75 ± 0.05b	7.90 ± 0.29b
	N_3SB_0	7.09 ± 0.20b	0.75 ± 0.05b	7.83 ± 0.25b
	N_0SB	2.40 ± 0.23d	0.29 ± 0.02d	2.69 ± 0.24d
	N1SB	7.16 ± 0.24b	0.73 ± 0.02b	7.90 ± 0.22b
	N_2SB	8.36 ± 0.15a	0.93 ± 0.03a	9.29 ± 0.13a
	N_3SB	8.24 ± 0.24a	0.90 ± 0.02a	9.15 ± 0.26a

Notes: Different letters indicate significant differences between treatments from 2022 to 2024 ($p < 0.05$, one−way ANOVA). The experimental treatments are as follows: N_0SB_0: no fertilization and biochar; N_1SB_0: 150 kg N ha^{-1} with no biochar; N_2SB_0: 225 kg N ha^{-1} with no biochar; N_3SB_0: N_3SB_0: 300 kg N ha^{-1} with no biochar; N_0SB: 40 t ha^{-1} biochar with no fertilization; N_1SB: 150 kg N ha^{-1} with 40 t ha^{-1} biochar; and N_2SB: 225 kg N ha^{-1} with 40 t ha^{-1} biochar; N_3SB: 300 kg N ha^{-1} with 40 t ha^{-1} biochar. Data shown are means±standard deviation (n = 3). Lowercase letters indicate differences between different treatments ($p < 0.05$).

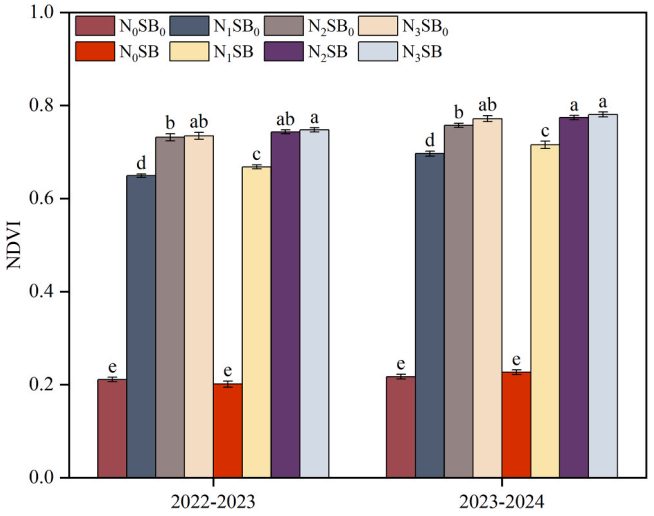


Fig. 3. The annual mean normalized difference vegetation index (NDVI) of turfgrass under different treatments. N_0SB_0: no N fertilization and biochar; N_1SB_0: 150 kg N ha^{-1} with no biochar; N_2SB_0: 225 kg N ha^{-1} with no biochar; N_3SB_0: 300 kg N ha^{-1} with no biochar; N_0SB: 40 t ha^{-1} biochar with no N fertilization; N_1SB: 150 kg N ha^{-1} with 40 t ha^{-1} biochar; and N_2SB: 225 kg N ha^{-1} with 40 t ha^{-1} biochar; N_3SB: 300 kg N ha^{-1} with 40 t ha^{-1} biochar. Error bars denote the standard deviation (n = 3). Lowercase letters indicate differences among the treatments ($p < 0.05$).

significantly enhanced turf growth and improved overall turf quality (Table 2, Fig. 3). Notably, the increase in turfgrass biomass and quality slowed as the N fertilizer rate increased. When N application exceeded 225 kg N ha^{-1}, no significant differences were observed in aboveground and belowground biomass or NDVI between N_2SB_0 and N_3SB_0 ($p > 0.05$). Under the same N gradient, biochar application significantly increased both aboveground and belowground biomass compared to non-biochar treatments, except in the N_0 treatment. The NDVI value for N_2SB was significantly higher than that of N_2SB_0 ($p < 0.05$), suggesting that biochar could improve turf quality (Fig. 3).

3.4. SOC content, storage, and sequestration rate

The average SOC content under biochar treatments was significantly higher than that under treatments without biochar (Fig. 4a). During the experiment, SOC content in biochar-amended treatments ranged from 8.57 to 11.99 g kg^{-1}, whereas treatments without biochar ranged from 3.60 to 4.93 g kg^{-1} (Fig. 4a). Additionally, from 2022 to 2024, biochar treatments exhibited significantly higher SOC storage compared to their non-biochar counterparts (Fig. 4b). Specifically, SOC storage in biochar-amended treatments (N_0SB, N_1SB, N_2SB, N_3SB) was 123.07 %–129.26 %, 121.68 %–139.04 %, 120.76 %–134.04 %, and 118.17 %–133.60 % higher, respectively, than in their non-biochar counterparts. By 2024, SOC storage (Mg C ha^{-1}) followed this order: N_0SB_0 (7.66) < N_1SB_0 (8.89) < N_3SB_0 (10.25) < N_2SB_0 (10.33) < N_0SB (17.55) < N_1SB (21.22) < N_3SB (23.94) < N_2SB (24.16) (Fig. 4b). Both N fertilizer and biochar enhanced the SOC sequestration rate (Fig. 4c). Under the same N fertilizer gradient, biochar amendment significantly improved the SOC sequestration rate. Among all treatments, N_0SB_0 had the lowest rate (-0.04 Mg C ha^{-1} yr^{-1}), while N_2SB showed the highest rate (3.36 Mg C ha^{-1} yr^{-1}) during the two-year period (Fig. 4c). These results indicated that neither unfertilized nor excessively fertilized conditions were conducive to SOC sequestration, whereas biochar application improves SOC stabilization.

3.5. Carbon footprint

The direct and indirect GHG emissions associated with turf management constituted the carbon sources of the CF (Fig. 5). In non-biochar, direct GHG emissions were the primary contributor to carbon sources, accounting for 69.87 %–74.25 % of the total, whereas biochar amendment reduced this proportion to 63.42 %–67.23 %. Compared to N_0SB_0, N_1SB_0 and N_2SB_0 reduced emissions by 10.37 % and 23.72 %, respectively, while N_3SB_0 increased emissions by 7.24 %. Under the same N fertilizer gradient, biochar amendment significantly reduced the CF by 13.02 %, 19.15 %, 24.94 %, and 23.34 % for N_0SB_0, N_1SB_0, N_2SB_0, and N_3SB_0, respectively, compared to treatments without biochar. The differences in CF were primarily driven by cumulative GHG emissions and SOC sequestration rates, with biochar treatments showing lower GHG emissions and higher sequestration rates than non-biochar treatments under equivalent N levels (Figs. 2d, 4c).

4. Discussion

4.1. Effects of N fertilization and biochar amendment on GHG emissions

N fertilization is a primary factor inducing soil N_2O emissions (Dutt and Tanwar, 2020). Our results showed that soil N_2O flux peaks

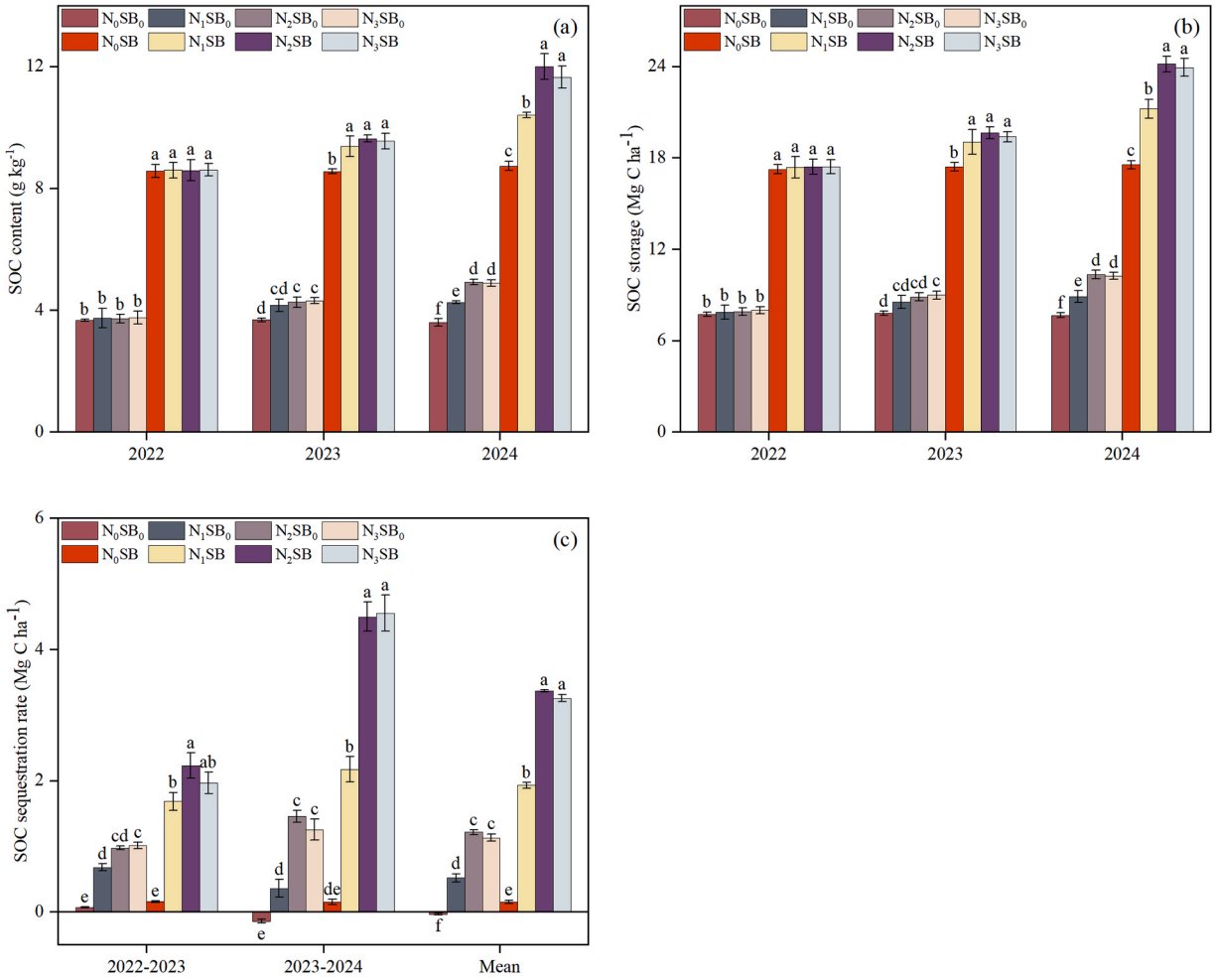


Fig. 4. The SOC content (g kg^{-1}) (a), SOC storage (Mg C ha^{-1}) (b), and SOC sequestration rate (Mg C ha^{-1}) (c) under different treatments from April 2022 to April 2024. N_0SB_0: no N fertilization and biochar; N_1SB_0: 150 kg N ha^{-1} with no biochar; N_2SB_0: 225 kg N ha^{-1} with no biochar; N_3SB_0: 300 kg N ha^{-1} with no biochar; N_0SB: 40 t ha^{-1} biochar with no N fertilization; N_1SB: 150 kg N ha^{-1} with 40 t ha^{-1} biochar; and N_2SB: 225 kg N ha^{-1} with 40 t ha^{-1} biochar; N_3SB: 300 kg N ha^{-1} with 40 t ha^{-1} biochar. Error bars denote the standard deviation (n = 3). Lowercase letters indicate differences among the treatments ($p < 0.05$).

occurred after each fertilization event (Fig. 1a). This phenomenon could be attributed to N fertilizer providing sufficient substrates for nitrification and denitrification reactions, thereby increasing soil N_2O emissions (Braker and Conrad, 2011; He et al., 2021). Consistent with previous studies, biochar amendment mitigated the stimulatory effect of N fertilization on N_2O emissions (Zheng et al., 2017; Wang et al., 2019b). Soil N_2O emission fluxes under biochar treatments were significantly lower than those under non-biochar treatments, with cumulative emissions reduced by 17.23–23.90 % (Fig. 2a). This reduction aligns with the reported range of N_2O emission reductions (6.89 %–29.13 %) achieved through biochar application (Liu et al., 2014). The suppression of soil N_2O emissions may result from biochar's high carbon-to-nitrogen (C/N) ratio and adsorption capacity, which limit substrate availability for nitrification and denitrification (Zhou et al., 2017). Additionally, biochar improved soil aeration by reducing bulk density (Fig. S1a) and increasing porosity, thereby inhibiting denitrification (Zhang et al., 2010, 2017; Zheng et al., 2017) and lowering N_2O emissions (Cayuela et al., 2014). Furthermore, in this study, the observed increase in soil pH following biochar addition (Fig. S1b) and the corresponding reduction in soil N_2O emissions (Fig. 2a) may be linked to the elevated abundance of the N_2O reductase gene *nosZ,* which is promoted by higher soil pH (Aamer et al., 2020).

The consistently positive soil CH_4 fluxes observed in this study (Fig. 1b) contrast with typical upland soils (Guo et al., 2024). This phenomenon can be attributed to frequent irrigation (50 events; Table S3), which maintained WFPS at an average of over 85 % (Fig. S2b). The high moisture levels created persistent anaerobic microsites, which enhanced the activity of CH_4-generating microorganisms, specifically acetoclastic methanogens and hydrogenotrophic archaea, thereby promoting soil CH_4 emissions (Grover et al., 2013; Conrad, 2007), as evidenced by the significant positive correlation between soil CH_4 fluxes and WFPS (Fig. S3). Kunnemann et al. (2023) also confirmed that irrigated park turf is a source of CH_4 when WFPS exceeded 75 %. Additionally, organic matter from turfgrass roots likely provided additional substrates for methanogenesis, further amplifying emissions (Table 2).

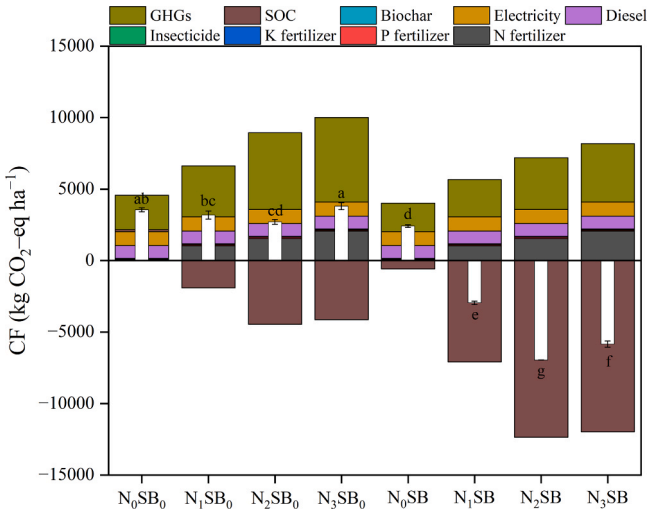


Fig. 5. The component of carbon footprints (CF) (kg CO_2-eq ha^{-1}) under different treatments from April 2022 to April 2024. N_0SB_0: no N fertilization and biochar; N_1SB_0: 150 kg N ha^{-1} with no biochar; N_2SB_0: 225 kg N ha^{-1} with no biochar; N_3SB_0: 300 kg N ha^{-1} with no biochar; N_0SB: 40 t ha^{-1} biochar with no N fertilization; N_1SB: 150 kg N ha^{-1} with 40 t ha^{-1} biochar; and N_2SB: 225 kg N ha^{-1} with 40 t ha^{-1} biochar; N_3SB: 300 kg N ha^{-1} with 40 t ha^{-1} biochar. Error bars denote the standard deviation (n = 3). Lowercase letters indicate differences among the treatments ($p < 0.05$).

Despite these high moisture conditions favoring soil CH_4 production, biochar application significantly reduced cumulative soil CH_4 emissions over the two-year period, enhancing carbon sequestration, compared with non-biochar treatments (Fig. 2b). These reductions may be attributed to changes in soil physical and microbial conditions induced by biochar. Biochar decreased soil bulk density (Fig. S1a), improving soil aeration and suppressing methanogenesis while enhancing methanotroph activity (Lu et al., 2019; Guo et al., 2022). The liming effect of biochar (Fig. S1b) and its rich nutrient content created more favorable conditions for methanotroph communities, as evidenced by higher *pmoA* gene abundance in biochar-amended soils (Jeffery et al., 2016). Furthermore, biochar adsorbed mineral nitrogen (NH_4^+-N and NO_3^--N), reducing competition for methane monooxygenase and minimizing toxic nitrite accumulation, thereby alleviating methanotroph inhibition and contributing to the reduced soil CH_4 emissions (Bodelier and Laanbroek, 2004).

Temperature is a significant factor influenced seasonal fluctuations of Re (Yang et al., 2022). Across all treatments in this study, the minimum values of Re were observed in winter, while the maximum levels occurred in summer (Fig. 1c). These results demonstrate the temporal variation pattern of Re, driven primarily by seasonal temperature changes (Fig. 1c, S2a), with elevated temperatures enhancing metabolic processes by mobilizing organic substrates that are biologically inactive at lower temperatures. This thermal activation accelerates microbial decomposition of soil organic carbon through intensified mineralization, leading to increased Re fluxes (Cui et al., 2021). However, no significant differences in soil temperature were observed among treatments, likely due to the consistently high soil moisture content resulting from frequent irrigation (Table S3), as water's high specific heat capacity buffers soil temperature against external thermal disturbances (Jha et al., 2017).

N fertilization increased Re fluxes (Fig. 1c). Previous studies suggest that CO_2 fluxes under N fertilization could exceed those from non-fertilized conditions by more than twofold (Huang et al., 2014). N fertilizer enhances microbial activity, promoting rapid SOC mineralization and thus facilitating CO_2 fluxes (Dossou-Yovo et al., 2016; Oladele and Adetunji, 2021). In contrast, biochar amendments showed no net effect on Re during the study ($p > 0.05$; Fig. 2c), attributable to two counteracting mechanisms. On the one hand, biochar promotes the formation of some macromolecular substances that are difficult to decompose by soil microorganisms (Oladele and Adetunji, 2021), reduces the utilization of organic carbon by microorganisms (Xia et al., 2024); microbial carbon-use efficiency is enhanced when SOC complexes on biochar surfaces and pores, leading to symbiosis and the preservation of soil microbes, nutrients, and substrate (Lehmann et al., 2011). On the other hand, biochar amendment mitigating the negative effects of nitrogen-induced soil acidification, significantly increasing soil pH (Fig. S1b), reducing aluminum toxicity, and enhancing microbial diversity and activity, which in turn increased microbial respiration (Li et al., 2018b; Shi et al., 2020). Elevated soil pH also promoted turfgrass growth and increased root biomass (Table 2), further facilitating autotrophic respiration (Davidson et al., 2004; Janssens et al., 2010). Additionally, biochar contains labile carbon that transiently increases soil respiration under microbial degradation, but this effect diminishes as biochar stabilizes (Smith et al., 2008). These opposing effects resulted in non-significant net changes in Re. Future studies should differentiate biochar's impacts on autotrophic versus heterotrophic respiration components in turf soils.

4.2. *Effects of N fertilization and biochar amendment on turf growth and quality*

The photosynthetic capacity and visual greenness of turfgrass are closely linked to plant nitrogen content (Hale et al., 2021). Fertilization, particularly with nitrogen, is a crucial factor in enhancing turf growth and quality (Johnson et al., 1987; Bilgili and

Acikgoz, 2011; Henning et al., 2013). In this study, N fertilization promoted turf growth, thereby increasing biomass and quality (Table 2, Fig. 3). However, no significant differences in biomass or quality were observed between N_2SB_0 and N_3SB_0, likely due to N application rates exceeding plant demand (Verdi et al., 2024). Under the same N fertilization rates, biochar amendments further enhanced turf growth, which agreed with previous studies (Montgomery, 2018; Hale et al., 2021). Notably, turf biomass and quality from 2023 to 2024 were higher than those from 2022 to 2023 (Table 2, Fig. 3), suggesting that biochar's benefits accumulate over time (Steiner et al., 2007; Spokas, 2013). Similar effects were reported for creeping bentgrass (*Agrostis stolonifera* L.), where biochar improved both aboveground and belowground growth and turf quality (Li et al., 2018a; Vaughn et al., 2018). These improvements can be attributed to biochar's capacity to improve soil conditions (Tian et al., 2019; Hale et al., 2021). First, biochar enhances nitrogen retention in the soil, reducing leaching (Singh et al., 2010; Crutchfield, 2016; Glab et al., 2016). Second, biochar application lowers soil bulk density and stabilizes soil aggregates, thereby enhancing N-cycling enzyme activities (Rogovska et al., 2011; Ibrahim et al., 2020). Third, biochar stimulates soil microbial abundance and activity, promoting root development and nutrient uptake (Raciti et al., 2008; Khan and Shea, 2013; Tian et al., 2019). Collectively, under reduced nitrogen conditions, biochar application is a sustainable strategy that significantly promotes turf growth and maintains quality.

4.3. Effects of N fertilization and biochar amendment on SOC storage

Increasing SOC sequestration through effective management is a widely recognized strategy for mitigating climate change (Bai et al., 2023). In this study, biochar amendment significantly increased SOC content (Fig. 4a). This enhancement can be attributed to biochar's dense aromatic and graphitic carbon structure, which resists microbial decomposition and reduces organic matter mineralization (Liu et al., 2021; Wang et al., 2022a). Liu et al. (2023) reported that biochar has greater carbon sequestration potential in alkaline soils with lower carbon saturation. Under the same nitrogen input, biochar application resulted in a higher SOC sequestration rate (Fig. 4c). Notably, biochar enhanced turfgrass root biomass (Table 2), leading to increased root litter input and external carbon input (Bolinder et al., 2007; Sohi et al., 2010), which subsequently increased the SOC sequestration rate and storage in the soil (Huang et al., 2022). Additionally, biochar provides a biodegradable organic carbon source for soil microbes, and regulating microbial growth, metabolic activity, and the accumulation of necromass through improved substrate efficiency (Liang et al., 2006; Chen et al., 2024a), aiding in the accumulation of SOC. These combined mechanisms—physical protection of carbon, enhanced root-derived inputs, and microbial-mediated stabilization—explain the significant SOC accumulation observed in biochar-amended treatments.

4.4. Effects of N fertilization and biochar amendment on GWP, NGWP, and CF

N fertilizer is a key driver of direct and indirect GHG emissions (Dachraoui and Sombrero, 2020). Our study revealed that increasing N application rates significantly elevated the GWP of turf systems compared to biochar application treatments (Fig. 2d). However, NGWP and CF did not follow this trend; instead, NGWP and CF generally showed a decreasing trend with increased N fertilizer application (Figs. 2e, 5). This discrepancy is primarily due to N fertilizer enhancing sequestration of SOC, which partially offset CO_2 equivalent emissions (Bai et al., 2023; Zhang et al., 2024a). Notably, compared with no biochar application, biochar amendment shifted CF from a "source" to a "sink" (except for N_0SB) (Fig. 5). This transition can be explained by two mechanisms: (i) biochar contributes additional stable carbon to the soil, reducing CF (Chen et al., 2024b); (ii) biochar decreased GWP (soil N_2O and CH_4) (Hu et al., 2024).

5. Conclusion

This two-year study evaluated the effects of nitrogen (N) fertilization and biochar amendment, on continuously monitored soil organic carbon (SOC) and greenhouse gas (GHG) emissions in a turf ecosystem. The results showed that biochar amendment reduced GHG emissions (N_2O: 17.34 %–32.90 %; CH_4: 18.15 %–25.56 %), net global warming potential (NGWP: 43.71 %–1046.17 %), and carbon footprint (CF: 32.01 %–356.66 %), while enhancing SOC sequestration (129.20 %–138.61 %), turfgrass total biomass (3.99 %–15.51 %), and normalized difference vegetation index (NDVI: −0.18 %–2.81 %) compared to N fertilization alone. Although soil moisture and temperature were confirmed as significant environmental drivers of GHG emissions, the absence of treatment-specific differences in these parameters may suggests that biochar's mitigation effects primarily stem from its inherent physicochemical properties (e.g., adsorption capacity and alkaline effect) rather than microclimate modification. Notably, compared to N_3SB, N_2SB significantly reduced soil N_2O emissions by 12.85 %, soil CH_4 emissions by 11.47 %, CF by 16.01 %, and NGWP by 9.82 %, while ecosystem respiration, SOC storage, turfgrass total biomass, and NDVI showed no significant differences between the two treatments. Therefore, N_2SB is a key practice for sustainable turf management. We emphasize that biochar has significant potential to promote turf growth and enable low-carbon maintenance. Although this study highlights biochar's potential under controlled conditions, its long-term efficacy and scalability require validation across diverse climates and turf management practices.

CRediT authorship contribution statement

Shen Songsong: Writing – original draft, Investigation, Formal analysis, Conceptualization. **Yang Zhimin:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition. **Pan Shang:** Writing – original draft, Investigation, Formal analysis. **Hu Jian:** Writing – review & editing, Formal analysis, Conceptualization. **Yu Jingjin:** Writing – review & editing, Methodology.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zhinmin Yang reports financial support was provided by Fundamental Research Funds for the Central Universities. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.eti.2025.104249.

Appendix B. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.eti.2025.104249.

Data availability

Data will be made available on request.

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